

# Emerging Directions in Deep Learning

## Curvatures

Mihai-Sorin Stupariu, 2025-2026

## Motivation

Curvatures for smooth curves and surfaces

Curvatures for discrete data types

# Use of curvatures in relationship to ML approaches

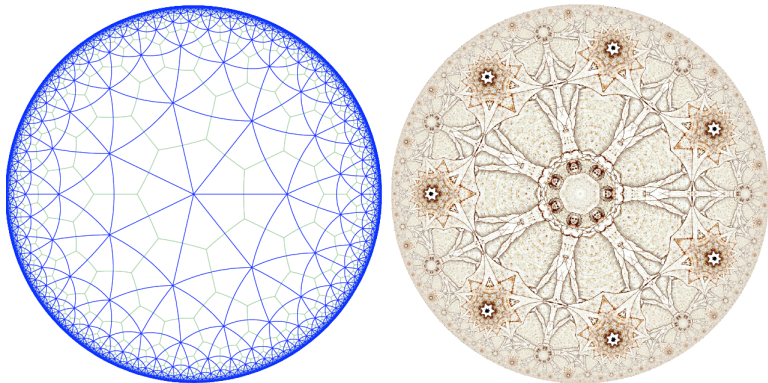
Examples:

Medical imaging

Facial expression

# Graph embeddings

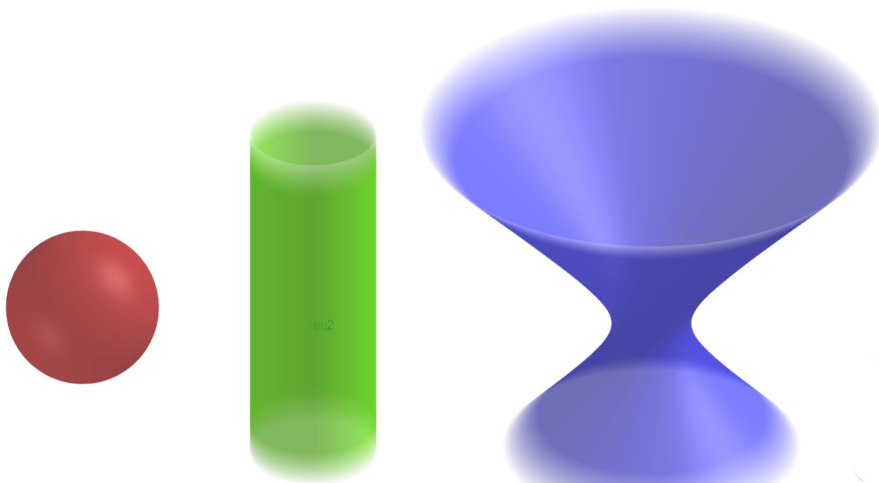
Papers: graph embedding in the hyperbolic space. Why?



Source: [Krioukov et al., 2010](#)

# Illustration

Is the geometry of these surfaces (2-dimensional manifolds) different?



## Curvature of a parametrized plane curve

Consider  $c : I \rightarrow \mathbb{R}^2$ ,  $c(t) = (x(t), y(t))$  a plane smooth parametrized curve. The curvature of  $c$  at  $c(t)$  is

$$\kappa(t) = \frac{|x'(t)y''(t) - x''(t)y'(t)|}{(x'(t)^2 + y'(t)^2)^{\frac{3}{2}}}.$$

Compute the curvature for  $c(t) = (\cos(t), \sin(t))$  and then for  $c_1(t) = (\cos(2t), \sin(2t))$ .

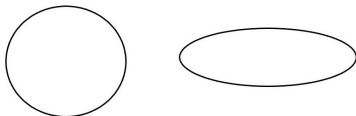
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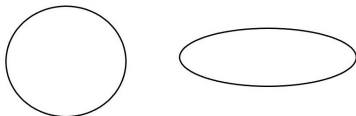
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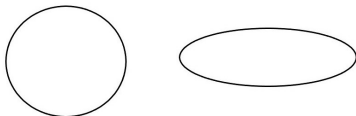
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## Gaussian and mean curvature of a parametrized surface

- ▶ Parameterization  $(u, v) \mapsto \varphi(u, v) \in \mathbb{R}^3$  of a parametrized surface (surface embedded in  $\mathbb{R}^3$ ). The **Gaussian curvature** at  $P = \varphi(u, v)$  can be computed as follows:

$$\kappa_G(P) = \frac{eg - f^2}{EG - F^2}.$$

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$$E = \left\| \frac{\partial \varphi}{\partial u} \right\|^2, \quad F = \left\langle \frac{\partial \varphi}{\partial u}, \frac{\partial \varphi}{\partial v} \right\rangle, \quad G = \left\| \frac{\partial \varphi}{\partial v} \right\|^2,$$

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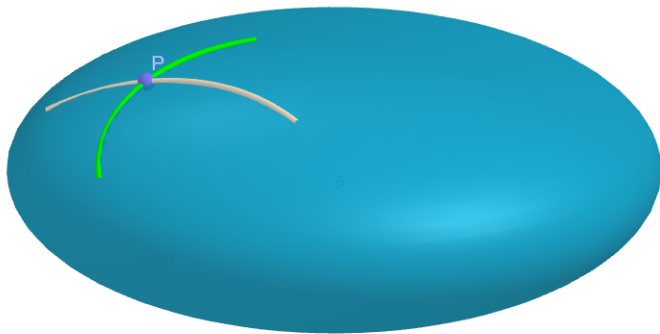
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- ▶ Try to compute it for the sphere  $(u, v) \mapsto (\cos(u) \cos(v), \cos(u) \sin(v), \sin(u))$  and for the cylinder  $(u, v) \mapsto (\cos(u), \sin(u), v)$ .

## Relationship to the local geometry of the surface

$\frac{\partial \varphi}{\partial u}, \frac{\partial \varphi}{\partial v}$  are spanning the tangent space at  $P$

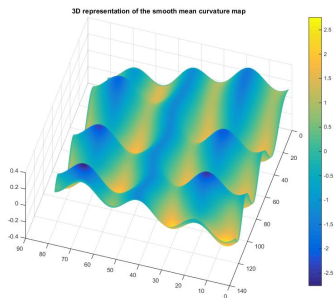
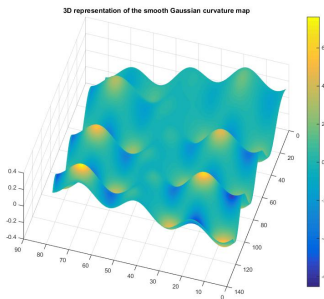
$N$  is the normal vector at  $P$



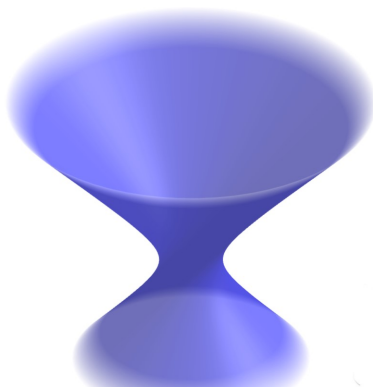
# Curvature maps for the Gaussian and for the mean curvature

The surface is given by

$$\varphi(u, v) = 0.2 \sin(u) \sin(v^{1/5}) - 0.1 \cos(5u), \quad u \in [1, 5], \quad v \in [1, 7].$$



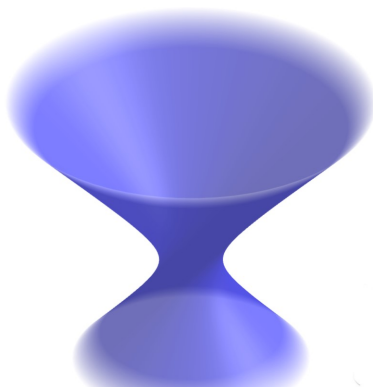
## Gaussian curvature for sphere, cylinder and hyperboloid

Positive  $\kappa_G$ Zero  $\kappa_G$ Negative  $\kappa_G$ 

(at any point, for each surface

speak about **elliptic** / **planar** / **hyperbolic** points)

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Could you imagine a **surface** that has all types of points?



# Instead of conclusion

TABLE I: Characteristic properties of Euclidean, spherical, and hyperbolic geometries. *Parallel lines* is the number of lines that are parallel to a line and that go through a point not belonging to this line, and  $\zeta = \sqrt{|K|}$ .

| Property                  | Euclidean                                                                         | Spherical                                                                         | Hyperbolic                                                                        |
|---------------------------|-----------------------------------------------------------------------------------|-----------------------------------------------------------------------------------|-----------------------------------------------------------------------------------|
| Curvature $K$             | 0                                                                                 | $> 0$                                                                             | $< 0$                                                                             |
| Parallel lines            | 1                                                                                 | 0                                                                                 | $\infty$                                                                          |
| Triangles are             | normal                                                                            | thick                                                                             | thin                                                                              |
| Shape of triangles        |  |  |  |
| Sum of $\triangle$ angles | $\pi$                                                                             | $> \pi$                                                                           | $< \pi$                                                                           |
| Circle length             | $2\pi r$                                                                          | $2\pi \sin \zeta r$                                                               | $2\pi \sinh \zeta r$                                                              |
| Disk area                 | $2\pi r^2/2$                                                                      | $2\pi(1 - \cos \zeta r)$                                                          | $2\pi(\cosh \zeta r - 1)$                                                         |

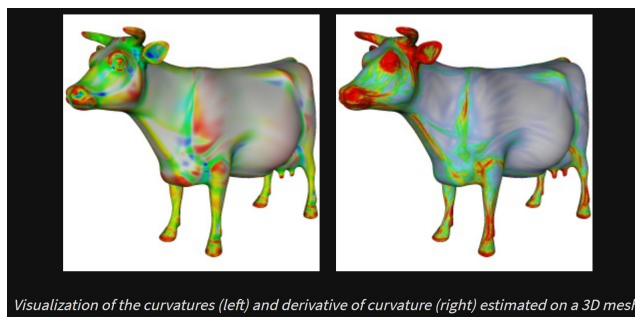
Source: [Krioukov et al., 2010](#)

In the hyperbolic model, the length of circles increases exponentially (similar to the number of nodes at a given distance with respect to the root in a  $b$ -ary tree).

# Discrete representations - triangle meshes

Instead of smooth surfaces, discrete representations are used: triangle / polygon meshes.

Curvatures maps for such meshes are widely used:

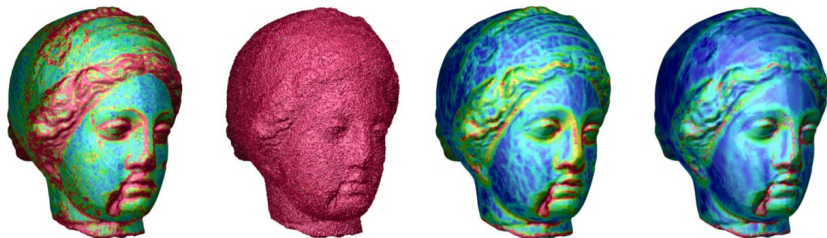


Source: [Rusinkiewicz, 2004](#)

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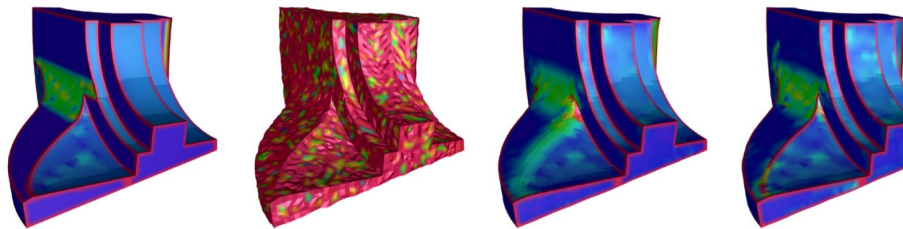


Source: [Hildebrandt and Polthier, 2004](#)

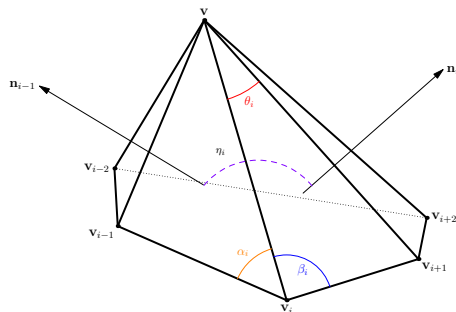
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Source: [Hildebrandt and Polthier, 2004](#)

Local structure. Notation: 1-ring  $\mathcal{N}_v$ 

Geometric elements around a vertex  $\mathbf{v}$ :

- ▶ Edges / faces incident to  $\mathbf{v}$  (or associated measures — lengths, areas).
- ▶ Angles  $(\theta_i)_i$  between edges incident to  $\mathbf{v}$ .
- ▶ Angles  $(\eta_i)_i$  between normals of faces incident to  $\mathbf{v}$ .
- ▶ Angles  $(\alpha_i)_i, (\beta_i)_i$  between edges of the 1-ring that are not incident to  $\mathbf{v}$ .

# Method: Gauss-Bonnet scheme (1) GB1

## ► Gaussian curvature at $\mathbf{v}$

$$K_{\mathbf{v}} = \frac{2\pi - \sum_{\mathbf{v}_i \in \mathcal{N}_{\mathbf{v}}} \theta_i}{\frac{1}{3}A}, \quad (1)$$

where  $2\pi - \sum_{\mathbf{v}_i \in \mathcal{N}_{\mathbf{v}}} \theta_i$  is the angular defect at  $\mathbf{v}$ , and  $A$  is the total area of the triangles in the 1-ring neighborhood of  $\mathbf{v}$

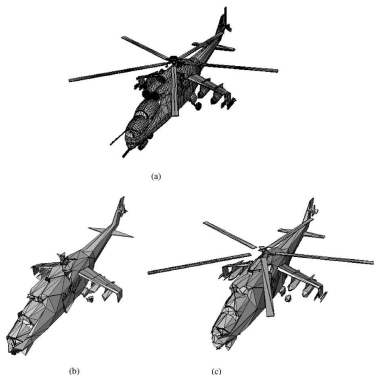
## ► Mean curvature at $\mathbf{v}$

$$H_{\mathbf{v}} = \frac{\frac{1}{4} \sum_{\mathbf{v}_i \in \mathcal{N}_{\mathbf{v}}} \|\overrightarrow{\mathbf{v}\mathbf{v}_i}\| \eta_i}{\frac{1}{3}A} \quad (2)$$

(measures the variation of the normals along the edges incident to  $\mathbf{v}$ )

## Method: Gauss-Bonnet scheme (1) GB1

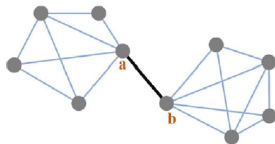
Used by [Dyn et al., 2001]; [Kim et al., 2002] for simplifying triangle meshes



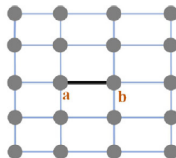
Helicopter model. (a) Original. (b), (c) Simplified versions. In (c) the discrete curvatures were used.

Source: [S.J. Kim, C.H. Kim, D. Levin, *Computers & Graphics*, 2002]

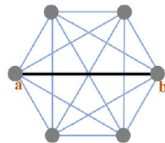
# Curvatures for graphs



(a) Negative Curvature



(b) Zero Curvature



(c) Positive Curvature

Source: [Li et al., 2022](#)